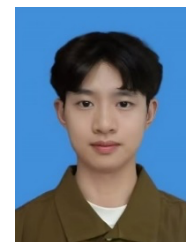


HaiTao Huang

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Forensic medicine is a profession that exists for the future



EDUCATION

Fudan University (FDU)

Master of Forensic Medicine

ZheJiang, CHINA

September 2026 – To date

- **Research Direction:** Forensic Genetics, Differential Extraction
- **Advisor:** Hua Chen

Westlake University (WU)

Visiting Researcher

ZheJiang, CHINA

July 2025 – July 2026

- **Research Direction:** Asteroid orbit prediction, PSF photometry, Rotational period determination
- **Advisor:** Shude Mao

University of Science and Technology of China (USTC)

(Purple Mountain Observatory, CAS)

Master of Astronomy

Nanjing, CHINA

September 2022 – May 2025

- **GPA:** 3.50/4.0
- **Main Courses:** Astronomical software (86), Deep space exploration progress, data processing and application (95), Advances in planetary sciences (90), Genetic mineralogy (85), Journal club (94)
- **Advisor:** Yunzhao Wu

Chengdu University of Technology (CDUT)

Bachelor of Geology (Earth Information Science and Technology)

SiChuan, CHINA

September 2018 – June 2022

- **GPA:** 3.50/5.0
- **Main Courses:** Sedimentary Petrology (94), Geographic Information Systems (92), Structural Geology (85), Metamorphic Petrology (84), Geological Remote Sensing (95), Geochemistry (88)

PUBLICATIONS

- [1] Zeng L, Li T, **Huang H T**, et al. Identifying Emeishan basalt by supervised learning with Landsat-5 and ASTER data[J]. *Frontiers in Earth Science*, 2023, 10: 1097778. [10.3389/feart.2022.1097778](https://doi.org/10.3389/feart.2022.1097778)
- [2] **Huang H T**, Wu Y Z. Review of Jupiter-Trojan asteroids research [J]. *Reviews of Geophysics and Planetary Physics*, 2024, 55(2): 175-183. [10.19975/j.dqyxx.2023-022](https://doi.org/10.19975/j.dqyxx.2023-022)
- [3] **Huang H T**, Wu Y Z, Wang L, et al. Quantitative Inversion of Minerals on Potentially Hazardous Asteroids[J]. *Journal of Deep Space Exploration*, 2024, 11(6):605-612. [10.15982/j.issn.2096-9287.2024.20240052](https://doi.org/10.15982/j.issn.2096-9287.2024.20240052)
- [4] Huang X T, **Huang H T**, Lei K Y. Quantitative Inversion of Siliceous Asteroid Minerals [J]. *Journal of Space Science and Experiment*, 2024, 1(04): 1-9. [10.19963/j.cnki.2097-4302.2024.04.001](https://doi.org/10.19963/j.cnki.2097-4302.2024.04.001)
- [5] Huang H T, Yang H J, Chung-Uk Lee, et al. Measuring Asteroid Rotation Periods Using the KMTNet Bulge Survey Data [J]. *Publications of the Astronomical Society of the Pacific*, 2026, 138(074401). [10.1088/1538-3873/ae7cb6](https://doi.org/10.1088/1538-3873/ae7cb6)

MANUSCRIPTS IN PREPARATION

- [1] **Huang H T**, Zhai R Y, Wu Y Z. Space Weathering Exposure Age Estimation of Potentially Hazardous Asteroids Using the Modified Gaussian Model
- [2] **Huang H T**, Qian Q Y, Yang H J, Mao S D, Gan T J. Rotational Period Determination of Asteroids Using KMTNet Data.

RESEARCH EXPERIENCE

Hunting for basalts in the Panxi region of Sichuan — from space!

Zeng L, *Huang H T*

Advisor: Ling Zeng

- **Overview:** This study employs widely used multispectral datasets—Landsat-5 and ASTER—to investigate the spatial distribution of basalts in the Panxi region of Sichuan Province.
- **Contributions:** Multispectral imagery covering the study area was acquired, and preprocessing procedures including radiometric calibration, atmospheric correction, geometric correction were conducted. Using the spectral information extracted from both Landsat-5 and ASTER datasets, followed by quantitative assessment of classification precision and accuracy for each model.

Figuring out what's really on asteroid surfaces — and how long they've been baking in space!

Huang H T, Wu Y Z, Zhai R Y, Wang P Y, Huang X T

Advisor: Yunzhao Wu

- **Overview:** This study applies the Modified Gaussian Model (MGM) and Cs Parameter to retrieve surface mineral compositions and exposure age of Near Earth Asteroids (including Potentially Hazardous Asteroids) and Main Belt Asteroids.
- **Contributions:** Spectral data of asteroids were acquired, preprocessed, and corrected for both phase - angle and thermal effects. The preliminary mineralogical estimates derived from the Gaffey model were incorporated as initial parameter constraints in the MGM fitting to achieve quantitative inversion of surface mineral assemblages. Furthermore, a robust empirical relationship was established between exposure age and MGM continuum slope.

Estimating how fast asteroids spin... with data that most people wouldn't expect you could use!

Huang H T, Qian Q Y, Yang H J, Mao S D, Gan T J,

Advisor: Shude Mao

- **Overview:** This study utilizes KMTNet observations to perform PSF photometry on asteroids passing through the field of view and subsequently determine their rotational lightcurve periods.
- **Contributions:** Accurate asteroid positions were calculated for fixed observational windows and sky regions. A long-timescale photometric dataset was constructed by performing PSF photometry across multiple sites and epochs. The extracted lightcurves were corrected for cross - station systematics, frame - to - frame differences, and phase - angle effects.

AWARDS & LANGUAGES

2025

- Outstanding Graduate of the University of Science and Technology of China (USTC)

2024

- National Graduate Scholarship (China)
- USTC First-Class Academic Scholarship
- “Triple-A Student” — School of Astronomy and Space Science, USTC (PMO, CAS)
- Excellent Chair — School of Astronomy and Space Science, USTC (PMO, CAS)
- Excellent Student Cadre — School of Astronomy and Space Science, USTC (PMO, CAS)

2022

- USTC First-Class Academic Scholarship
- Outstanding Graduate of Sichuan Province
- Outstanding Graduate of Chengdu University of Technology (CDUT)

Languages: Mandarin Chinese (native), Cantonese, English (CET6: 483; CET4: 524)